WORKED SOLUTION

STEP I 1993, Question 16

Probability and Statistics • Geometrical Probability

STEP I 1993, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q16

This question has a full guided treatment in *Beyond STEP*, Section 12.20, *Buffon's Needle*: https://beyondstep.keithmcnulty.org/ch12/buffon.html

The key idea

This is Buffon's needle. Whether the pin crosses the nearest line depends only on how far its tip reaches *towards* that line, and that reach is the projection of a half-pin onto the perpendicular direction. Fix the distance of the centre first, find the conditional probability from the angle alone, then average over the distance. The integral in the first part is exactly the one that this averaging produces, which is why the question starts there.

Solution

The integral

Put $y = \cos^{-1} t$, so $t = \cos y$ and $\mathrm{d}t = -\sin y \,\mathrm{d}y$. When $t = 0$, $y = \frac{\pi}{2}$, and when $t = 1$, $y = 0$. Hence

$$\int_0^1 \cos^{-1} t \,\mathrm{d}t = \int_{\pi/2}^0 y\,(-\sin y)\,\mathrm{d}y = \int_0^{\pi/2} y \sin y \,\mathrm{d}y.$$

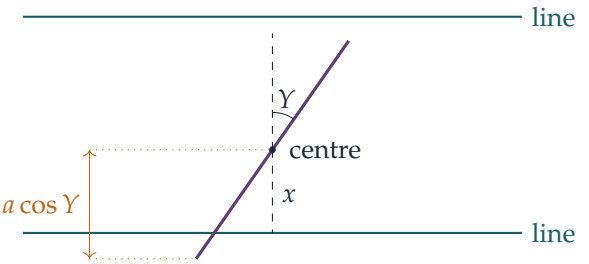
Integrating by parts with $u = y$ and $\mathrm{d}v = \sin y \,\mathrm{d}y$,

$$\begin{aligned}\int_0^{\pi/2} y \sin y \,\mathrm{d}y &= \Big[-y\cos y\Big]_0^{\pi/2} + \int_0^{\pi/2} \cos y \,\mathrm{d}y \\ &= 0 + \Big[\sin y\Big]_0^{\pi/2} = 1,\end{aligned}$$

as required.

(Or otherwise: the integral is the area between $y = \cos^{-1} t$ and the axes, which is the same region as the area under $t = \cos y$ for $0 \le y \le \frac{\pi}{2}$, namely $\int_0^{\pi/2} \cos y \,\mathrm{d}y = 1$.)

The probability given $X = x$



Each half of the pin has length a and makes angle Y with the perpendicular to the lines, so the ends of the pin lie a perpendicular distance $a\cos Y$ on either side of the centre. The nearest line is at distance x from the centre, so the pin crosses it exactly when

$$a\cos Y \;\geq\; x.$$

(Equality, where the tip just touches the line, has probability zero, so it doesn't matter whether we count it.)

Ask yourself How do we turn $a\cos Y \geq x$ into a condition on Y alone?

Since cos is decreasing on $[0, \frac{\pi}{2}]$, it depends on whether x/a is attainable.

- If $x \geq a$, then $a\cos Y \leq a \leq x$ with equality only when $Y = 0$ and $x = a$, so the probability is 0.
- If $0 \leq x < a$, then $\cos Y \geq x/a$ is equivalent to $Y \leq \cos^{-1}(x/a)$, where $\cos^{-1}(x/a) \in (0, \frac{\pi}{2}]$. As Y is uniform on $[0, \frac{\pi}{2}]$,

$$\mathrm{P}(\text{crosses} \mid X = x) = \frac{\cos^{-1}(x/a)}{\pi/2} = \frac{2}{\pi}\cos^{-1}\frac{x}{a}.$$

Here we used the independence of X and Y: knowing $X = x$ leaves Y uniform on $[0, \frac{\pi}{2}]$.

The probability for a general throw

Now let $a < b$. The density of X is $1/b$ on $[0, b]$. Because $a < b$, the interval $[0, a)$ where crossing is possible lies inside $[0, b]$, and the conditional probability is zero on $[a, b]$. By the law of total probability,

$$\mathrm{P}(\text{crosses}) = \int_0^b \mathrm{P}(\text{crosses} \mid X = x)\,\frac{1}{b}\,\mathrm{d}x = \frac{1}{b}\int_0^a \frac{2}{\pi}\cos^{-1}\frac{x}{a}\,\mathrm{d}x.$$

Substituting $x = at$, $\mathrm{d}x = a\,\mathrm{d}t$, and using the first part,

$$\mathrm{P}(\text{crosses}) = \frac{2a}{\pi b}\int_0^1 \cos^{-1} t\,\mathrm{d}t = \frac{2a}{\pi b},$$

as required. Since $a < b$, the pin is shorter than the gap $2b$ and can never reach a second line, so "crosses a line" and "crosses the nearest line" are the same event.

Checks

- As $a \to b$ the answer tends to $2/\pi \approx 0.637$, which is less than 1, as it must be: a pin as long as the gap still often lies nearly parallel to the lines.
- The answer is proportional to a: doubling a short pin doubles the chance of a crossing, which matches the fact that the expected number of crossings is linear in the length.
- A simulation of the actual geometry (centre uniform across a strip, angle uniform) with $a = 1$, $b = 2$ gave 0.3183 over 4×10^5 throws, against $1/\pi \approx 0.31831$. Midpoint quadrature gave $\int_0^1 \cos^{-1} t \, \mathrm{d}t = 1.0000000$.

Common slips

- Using $a \sin Y$ instead of $a \cos Y$: here Y is measured from the *perpendicular* to the lines, not from the lines themselves.
- Integrating the conditional probability over all of $[0, b]$ with the formula $\frac{2}{\pi} \cos^{-1}(x/a)$, which is undefined for $x > a$; the conditional probability is 0 there.
- Forgetting the density $1/b$ when averaging over X.

Beyond STEP

Buffon posed this in the eighteenth century, and it is the classic example of *geometric probability*: probabilities computed as areas in a space of parameters, here the rectangle $[0, b] \times [0, \frac{\pi}{2}]$. Since the answer involves π, throwing pins gives a (very slow) Monte Carlo estimate of π. A beautiful extension, "Buffon's noodle", shows by linearity of expectation that the expected number of crossings for a curve of any shape depends only on its length. Grimmett and Welsh, *Probability: An Introduction*, covers continuous joint distributions and conditioning of this kind.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.